

Datagram - Key Points

Definition:

A datagram is an independent packet of data sent over a connectionless network. Each one carries enough information to reach its destination without needing prior setup.

Protocol:

Used mainly in UDP (User Datagram Protocol), which is part of the Internet Protocol suite.

Header Info:

Each datagram includes source and destination addresses, length, and checksum.

Connectionless Communication:

No session or handshake between sender and receiver.

Independent Routing:

Each datagram can take a different path through the network.

Unreliable Delivery:

No guarantee that datagrams will arrive, arrive in order, or arrive only once.

Faster Transmission:

No connection setup or acknowledgment — makes it lightweight and quick.

No Congestion Control:

Network traffic can drop packets if overloaded.

Examples of Use:

UDP applications like DNS, online games, live streaming, and VoIP.

Analogy:

Like sending individual postcards — each one has its own address, travels separately, and may or may not reach the destination.